# EP05 — Why do computers only understand 0s and 1s?

HINGLISH (Roman)

## 01 Title

Computers sirf **0s aur 1s** hi kyun samajhte hain? Welcome to Katixo KhojLab. Aapke phone ki har photo, har song, har video — andar se bas do hi digits hain. Aaj pata karte hain aisa kyun.

> **[CUE]** Title pe ~2s ruko, calm confident tone, phir aage badho.

## 02 The Hook

Apne phone ko dekho. Har photo, har song, har video, har WhatsApp message — **sab kuch**, andar se, bas zero aur one hai. Aur yahan twist hai: hum insaan **das** digits se ginte hain, zero se nine. Toh ab tak ki sabse powerful machine sirf **do** hi kyun use karti hai? Ruko paanch minute — ye poori mystery khatam ho jayegi.

> **[CUE]** Yahan energy high — ye hook hai. "har photo, har song" pe phone uthao.

## 03 Aaj ka sawaal

Toh aaj ka ek sawaal: computer sirf **do** symbols kyun use karta hai — aur sirf do symbols se **sab kuch** kaise store ho sakta hai? Chalo pata karte hain.

> **[CUE]** Thoda slow. Sawaal ko land hone do, phir aage.

## 04 Idea 1 — Switches

Idea number one: computer billions of microscopic **switches** se bana hota hai. Har switch ko **transistor** kehte hain. Aur ek switch sirf do positions mein ho sakta hai — **off**, ya **on**. Bas. Hum un do positions ko naam de dete hain: off ko bolte hain **zero**, on ko bolte hain **one**. Yahi hai computer ki poori alphabet — sirf do letters.

> **[CUE]** "off... on" pe light switch ka mime karo. Pehle 0 reveal, phir 1.

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## 05 Idea 2 — Two = reliable

Ab bada sawaal — sirf **do** kyun? Das kyun nahi, hamari tarah? Yahan key insight hai. Do states ko alag pehchaanna bahut **aasaan** hai — bilkul light switch ki tarah, jo clearly on ya clearly off hota hai, jo chip ke andar matlab high voltage ya low voltage. Ab socho das alag-alag voltage levels thoonse hue hain. Wo six tha ya seven? Thodi si noise aayi aur **galti** ho gayi. Toh do states matlab simple, simple matlab **reliable**, aur reliable matlab fast. Bas yahi poora reason hai.

> **[CUE]** "reliable" pe zor — ye video ka sabse important idea hai.

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## 06 Idea 3 — Combine bits

"Par do options toh bahut kam hain!" — sahi? Akela ek switch sirf zero ya one bol sakta hai. Trick ye hai ki aap unhe **line mein laga do**. Har zero ya one ko **bit** kehte hain. Chaar bits ko ek row mein rakho — jaise one-zero-one-zero — aur place value se padho, toh wo aath plus do, matlab **das**. Aath bits ko jodo toh ek **byte** banta hai — yaani do sau chappan alag combinations. Itne bits jodo, aur aap **koi bhi** number represent kar sakte ho.

> **[CUE]** 8-4-2-1 columns reveal karo, phir "= 10" result. Dheere padho.

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## 07 Idea 4 — Everything becomes numbers

Theek hai — par photo toh number nahi hai, song toh number nahi hai. Toh yahan final move hai. **Sab kuch** pehle numbers mein badal jaata hai. Letters ko ek number code milta hai — wo hai `ASCII` ya `Unicode`. Colours numbers ban jaate hain — har dot ke liye ek red, green aur blue value. Sound numbers ban jaata hai — har second mein hazaaron chhote samples. Aur jab sab numbers ban gaye, toh **har** number binary ban jaata hai. Toh ek bahut hi simple system literally sab kuch store karta hai.

> **[CUE]** letters → colours → sound rows ek-ek karke reveal karo jab list karo.

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## 08 Why it matters

Aur yahi sabse khoobsurat payoff hai. Isiliye **"digital"** itna reliable aur itna sasta hai. Do clean states — on ya off — isi wajah se aapka computer data padhte waqt almost kabhi **galti nahi** karta. Aur kyunki har switch itna simple hai, use chhota karke ek hi chip pe billions switches pack kiye jaa sakte hain — tiny, sasta, aur fast. Poori digital duniya isi ek idea pe khadi hai.

> **[CUE]** Confident, thoda awed tone — ye "wah, kitna elegant" wala beat hai.

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## 09 Common myth

Aur isse ek bada myth clear ho jaata hai. Log sochte hain binary koi weird, mysterious **secret code** hai jo sirf computers samajhte hain. Aisa bilkul nahi hai. Ye bas das ki jagah **do symbols se ginti** hai — aur do hone ki ek hi wajah hai ki hardware mein physically do states hain: on-off switches. Isme kuch bhi magical nahi hai.

> **[CUE]** Myth pe ek beat ke baad green "Reality" card reveal karo.

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**10** Recap

Quick recap. Ek: computer bas billions of **on-off switches** hai — toh iski poori alphabet zero aur one hai. Do: do states **reliable** hain, aur kaafi saare jodne se aap **koi bhi** data represent kar sakte ho — ek number, ek letter, ek photo, ek song.

> **[CUE]** Crisp aur clear — ye do lines log yaad rakhein.

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**11** CTA

Bas itna hi — zeros aur ones bas aise switches hain jin par aap **bharosa** kar sakte ho. Agar finally binary aapko click ho gaya, toh Katixo KhojLab ko subscribe karo. Agla sawaal: computer in saare bits se ek **photo** store kaise karta hai? Main dikhaunga. Milte hain agle video mein.

> **[CUE]** Warm, friendly. Subscribe line pe awaaz mein smile.

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